

NutrioBox Dispatch — New Operating Rules & Architecture

Agreed specification for the gun (handheld) app and the PACT worker. Architecture: server-driven (Path 1). Submit reconciliation: per batch. Compiled 19 Sep 2026 — v1, for confirmation before build.

Guiding principle

PACT is the single source of truth. Every check, compare and stock read is taken from PACT live — never from the device's cached copy. The device only scans and displays; PACT decides what is owed, what is delivered, and what may still be posted. PACT's per-line pending cap is the permanent backstop: nothing can ever over-dispatch.

The rules

Rule 1 — Compare & sync from PACT, never the device

- **1a. Auto-post check/compare from PACT.** The delta (what is still owed vs already delivered) is computed from PACT's live pending, not the local ledger. **Status: DONE — deployed on the worker today.**
- **1b. Inventory/stock sync & compare from PACT.** Stock levels used for every decision come from a fresh PACT read, not the device snapshot.

Rule 2 — Auto-post error handling (show, wait, retry without the bad items)

- On an auto-post error, the gun shows the **exact PACT error text** and **which item(s)** it came from.
- That error stays visible for **10 seconds**.
- The auto-post then **retries automatically, leaving out the error item(s)** so every good item still posts and only the flagged items wait.
- **Every later cycle keeps skipping the flagged item(s)** — showing their exact PACT reason — until someone fixes and re-scans them. Nothing else is ever blocked by one bad item.

Rule 3 — Batchwise final submit

When Submit is pressed, for **each batch**: **post = handheld-scanned(batch) – PACT-delivered(batch)**. The worker fetches PACT's already-delivered quantity **per batch** (from the posted voucher lines for that OMR) and posts exactly the remainder. Batch-level, not product-level.

Rule 4 / 5 / 6 — Timings

- **Rule 4:** Auto-post interval is **15 minutes** (was 10).
- **Rule 5:** Every **13 minutes**, inventory is synced from PACT for **all items in that specific SO**.
- **Rule 6:** Auto-post runs at the **15-minute** mark — so a fresh 13-minute stock sync always lands ~2 minutes before each post.

Architecture — server-driven (Path 1)

One flash of the new gun build sets up the pieces that must live on the device; after that, the worker owns all scheduling and the timings become server config you can change without ever flashing again.

Lives on the GUN (this one flash)	Lives on the WORKER (server, tunable without flashing)
Continuously publish the live scan draft per SO to the server (so the worker always has the current scanned packs).	Per active SO: sync inventory from PACT every 13 min; auto-post the PACT-truth delta every 15 min.
Show the exact PACT error + offending items for 10s; reflect skipped/ retrying state.	On a PACT error: record the exact reason + error items, skip them, post the rest, keep skipping until fixed.
Batchwise Submit button (flags the SO for a final post).	Batchwise submit: read PACT delivered per batch, post scanned – delivered per batch.
Local 10-min auto-post timer is removed (worker drives it now).	Holds the 13/15-min timings and all compare-from-PACT logic as config.

Data contract (new, to be added on the server)

- **Live scan draft** — per SO: OMR, outlet, the current scanned barcodes, a completed flag and a submit-requested flag, with an updated-at timestamp. The gun upserts this on every scan change; the worker reads it to know what to post.
- **Post results** — per SO per cycle: what posted, what was skipped, the exact PACT error, the error items, the voucher number and status. The worker writes it; the gun polls it to show the 10-second error and the live status.
- **Config** — the 13 and 15 minute intervals and related settings, changeable server-side with no flash.

Build phases

- **Phase 1 — Worker foundation:** the data-contract stores, a batch-level PACT delivered read, and the timing config.
- **Phase 2 — Worker scheduler:** per active SO, run the 13-min sync and 15-min post from the live scan draft, with the error-skip and batchwise-submit logic.
- **Phase 3 — Gun app (the one flash):** publish the live scan draft; the 10-second error UI + skip-retry display; the batchwise Submit; remove the local auto-post timer.
- **Phase 4 — Integration test** on a single live SO, then roll out.

Until the flash, the current app-driven auto-post keeps running with the PACT-truth fix already live — so there is no regression while the new build is prepared.

Reference for the NutrioBox / Sukhna Foods dispatch build. Rule 1 is live; Rules 2–6 land with Phases 1–3 and the one gun-build flash.